

SciREX: Scientific Relation Extraction

Midterm Presentation

Team #3

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July 14, 2025

Summer Semester 2025

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Overview

Research Topic

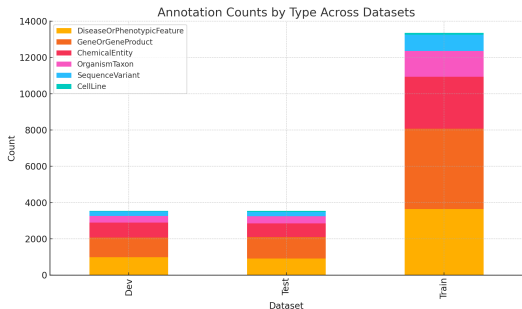
- Tackling information overload in biomedical literature as a generative information extraction task.
- Focus on automated relation extraction (RE) from scientific texts.
- Use BioRED dataset with rich biomedical entity/relation annotations as a unified benchmark.
- Goal: Scalable, automated scientific knowledge curation using LLMs.

Objectives

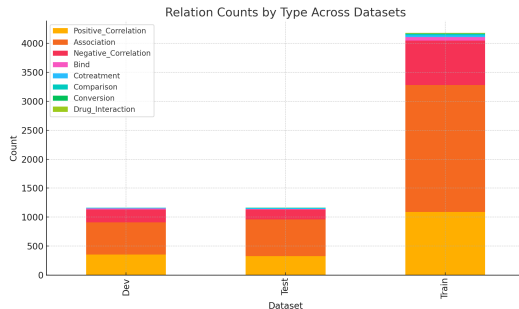
- Evaluate classification-based, LLMs instruction alignment as QA, and generative RE methods.
- Turn BioRED into a robust benchmark for comparison.
- Analyze both quantitative and qualitative performance.
- Explore generalization across domains and tasks.

Dataset Implications

- Class imbalance: rare entities and relations.
- Large training split allows diverse model training.
- As shown below, the annotation and relation distributions vary across splits.



(a) Annotation Counts



(b) Relation Counts

Figure: Entity and relation distributions across datasets.

Code: GitHub Repository

BioBERT

BioBERT for Relation Extraction

- Fine-tuned BioBERT with classification head on BioRED
- Uses sentence, entity1, entity2 $\rightarrow$ predict relation
- Fully reproducible pipeline (config-based)
- Supports full and mini datasets (via CLI flag)

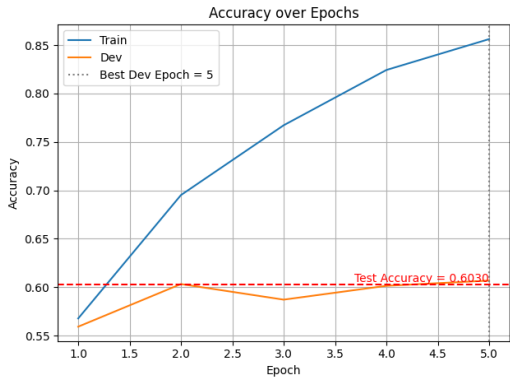
Training & Evaluation Pipeline

- Preprocessing: Converts `.BioC.JSON` files to `.tsv` format.
- Supports both full and mini datasets via `--mini` CLI flag.
- Logs step-wise loss and accuracy; epoch-level macro precision, recall, F1.
- Evaluation reports saved for each epoch (dev) and final test.
- Plotting via `--gen-result-plots`: train/dev/test metrics over epochs.

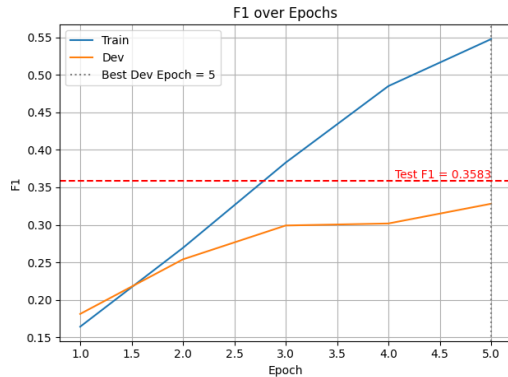
Training Performance & Evaluation

- Accuracy and F1 scores improved consistently over 5 epochs.
- Best epoch was the final one → training not yet saturated.
- Dev and test metrics aligned: no overfitting observed.
- High performance on frequent classes (e.g., Association).
- Zero F1 on rare classes (e.g., Bind, Drug-Interaction).

Metrics over Epochs



(a) Accuracy over Epochs



(b) F1 score over Epochs

Figure: Metrics over the epochs run

Insights So Far

- Model shows clear learning trajectory — can benefit from more epochs.
- Setup is robust: all logs, reports, and plots are automatically generated.
- Class imbalance is currently the major performance bottleneck.

Next Steps

- Extend training to 10–15 epochs with early stopping.
- Experiment with hyperparameter optimisation, model architecture reworking, class weighting and data augmentation.
- Add confusion matrix and per-class trend plots.

QA4RE

Question Answering for Relation Extraction

Why QA4RE?

- RE makes up less than 1% of all tasks used to train LLM models.
- QA4RE aligns RE with question answering, a predominant task in instruction-tuning.

How to implement QA4RE?

- Reframe the RE task as a multiple-choice question with zero-shot instruction
- LLM selects the right relation.
- Underrepresented task of RE → common QA format

QA4RE vs. Vanilla Prompting

Vanilla (Standard) Prompts:

- Simple classification: "Given the passage and entities, classify the relationship."
- Presents entities and a list of possible relation labels.
- Model must output the relation type directly (e.g., "Association", "No Relation").

QA4RE (Question Answering for Relation Extraction):

- Prompt is restructured as a question-answering (QA) multiple-choice task.
- Model chooses from natural language options (A, B, C, D), each describing a possible relation.
- Emphasizes inference and comprehension rather than label memorization.

QA4RE vs. Vanilla Prompting

Vanilla RE

Given a sentence, and two entities within the sentence, classify the relationship between the two entities based on the provided sentence. All possible relationships are listed below:

- per:city_of_birth
- per:city_of_death
- per:cities_of_residence
- no_relation

Sentence: **Wearing jeans and a white blouse, Amanda Knox of Seattle is being cross-examined by prosecutors.**

Entity 1 : **Amanda Knox**

Entity 2 : **Seattle**

Relationship: **per:city_of_birth** ❌

QA4RE

Determine which option can be inferred from the given sentence.

Sentence: **Wearing jeans and a white blouse, Amanda Knox of Seattle is being cross-examined by prosecutors.**

Options:

- A. Amanda Knox was born in the city Seattle
- B. Amanda Knox died in the city Seattle
- C. Amanda Knox lives in the city Seattle
- D. Amanda Knox has no known relations to Seattle

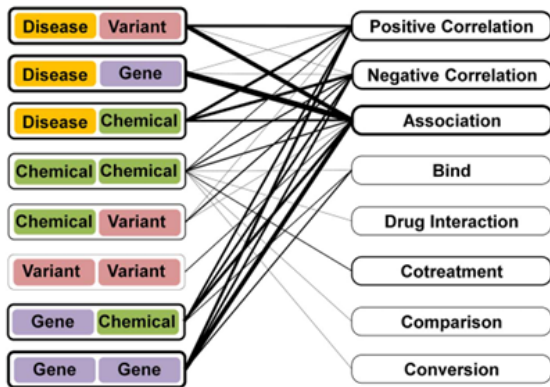
Which option can be inferred from the given sentence?

Option: **C.** ✅

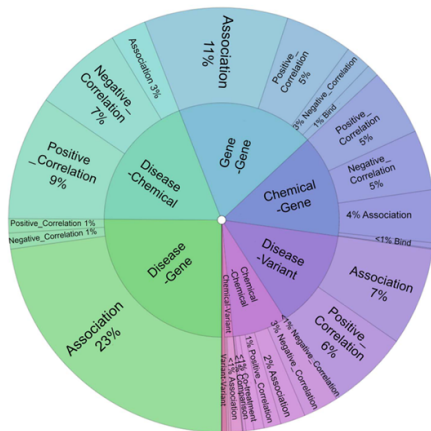
From BioRED JSON to QA4RE Prompts

- **BioRED JSON:** documents with entities, relations, and passage text.
- Options are full sentences in natural language, e.g.,
A. Disease and Gene are positively correlated.
- **Example (QA4RE-style prompt):**
Determine which option can be inferred from the given passage.
Passage: [Biomedical text]
Options:
A. Relation 1
B. Relation 2
...
H. No relation
Which option can be inferred? Option:
- Only biologically valid entity pairings are included to permit certain relation types.

From BioRED JSON to QA4RE Prompts



(a) Relations constraints between entities in BioRED



(b) Relations distribution for entity pairs

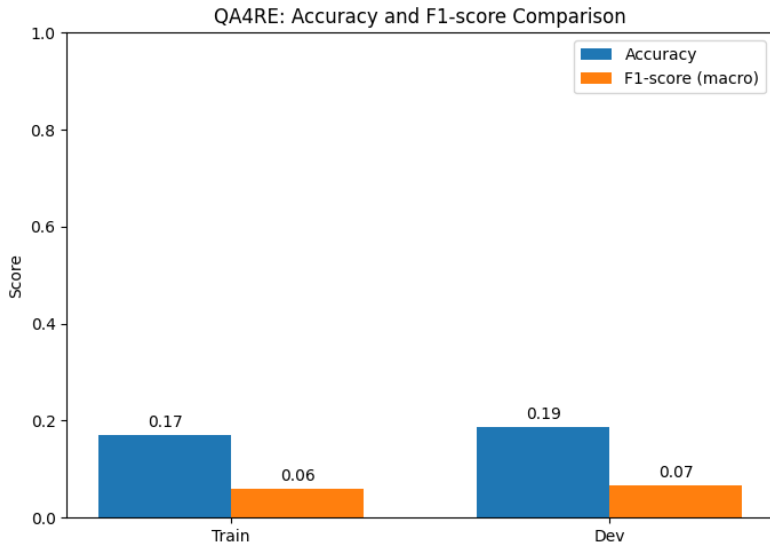
Model Configuration

- **Model Used:** google/flan-t5-small (~ 80 M parameters), Huggingface Transformers
- **Prediction:**
 - Model generates a short text (e.g., “A”, “B”, etc.).
 - Output is post-processed to extract a valid option or “INVALID” if no valid option is detected.
- **Type Constraints:** Only valid relations (options) are offered per entity type-pair.
- **Zero-shot**

Output Analysis and Metrics

- **Dev dataset:**
 - Size: 1284
 - Accuracy: 18.61%
 - F1 (weighted: scaled per class distribution): 0.2057
 - **F1 (macro): 0.07**
 - Invalid predictions: 297 (23.13%)
- **Train dataset:**
 - Size: 4497
 - Accuracy: 17.08%
 - F1 (weighted): 0.1877
 - **F1 (macro): 0.06**
 - Invalid predictions: 695 (15.45%)
- **Observations:**
 - Model performance is lower (possibly cause the model used is small).
 - Notable percentage of invalid predictions (even after post-processing)

Results



Key Takeaways and Next Steps

- **Small LLMs (e.g., FLAN-T5-small)** struggle to predict biomedical RE on passage/sentence level, yielding low accuracy and high invalid rates 15-23%.
- **Next steps improvement:**
 - Use larger, more instruction-tuned models.
 - Further tune prompts and improve constraints on output decoding.
 - Test with few-shots

Scifive

Why Use SciFive?

- SciFive is trained on biomedical text (PubMed).
- It understands biomedical language better than general models.
- Based on the T5 model, well-suited for text-to-text tasks.
- Ideal for generating relation labels in natural language.

Data Preparation

- Extracted sentence-level context and entity mentions from BioRED.
- Paired entities to form relation candidates.
- Example: “aspirin treats inflammation” → Target: “treats”

Model Training

- Fine-tuned SciFive on formatted input-output pairs.
- Optimized model using training + validation splits.
- Saved checkpoints regularly to monitor learning progress.

Prediction

- Applied trained model to new sentences.
- Generated relation types in natural language.
- Compared outputs to held-out annotations.

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